String Problems (1–50)

1. Write a program to find and display the total number of characters present in a given string.
2. Write a program to print each character of a string on a separate line.
3. Write a program to convert all characters of a string to uppercase.
4. Write a program to convert all characters of a string to lowercase.
5. Write a program to reverse a given string and display the result.
6. Write a program to check whether a given string is a palindrome or not.
7. Write a program to count the total number of vowels in a string.
8. Write a program to count the total number of consonants in a string.
9. Write a program to count how many digits are present in a string.
10. Write a program to count the number of spaces in a string.
11. Write a program to count the number of special characters in a string.
12. Write a program to count the total number of words in a sentence.
13. Write a program to find how many times a particular character occurs in a string.
14. Write a program to find the frequency of every character in a string.
15. Write a program to identify and display duplicate characters in a string.
16. Write a program to remove duplicate characters from a string while keeping the first occurrence.
17. Write a program to replace all spaces in a string with a specific character such as '-' or '_'.
18. Write a program to remove all spaces from a string.
19. Write a program to check whether two strings are exactly equal.
20. Write a program to compare two strings and determine which one comes first alphabetically.
21. Write a program to join two strings into a single string.
22. Write a program to check whether a string starts with a given prefix.
23. Write a program to check whether a string ends with a given suffix.

24. Write a program to check whether a string contains a particular word or substring.
25. Write a program to find the position of the first occurrence of a character in a string.
26. Write a program to find the position of the last occurrence of a character in a string.
27. Write a program to extract a substring from a given string using specified start and end positions.
28. Write a program to split a sentence into individual words and display them.
29. Write a program to join multiple strings stored in an array into a single sentence.
30. Write a program to remove all vowels from a string.
31. Write a program to remove all consonants from a string while keeping vowels.
32. Write a program to convert uppercase letters to lowercase and lowercase letters to uppercase.
33. Write a program to capitalize only the first character of a string.
34. Write a program to capitalize the first letter of every word in a sentence.
35. Write a program to display the ASCII value of each character in a string.
36. Write a program to count the number of uppercase and lowercase letters separately.
37. Write a program to check whether a string contains only numeric characters.
38. Write a program to check whether a string contains only alphabetic characters.
39. Write a program to check whether a string contains only letters and digits.
40. Write a program to reverse every word in a sentence without changing the word order.
41. Write a program to reverse the order of words in a sentence.
42. Write a program to find the longest word in a sentence.
43. Write a program to find the shortest word in a sentence.
44. Write a program to count how many times a given substring appears in a string.
45. Write a program to display the first character of a string.
46. Write a program to display the last character of a string.

47. Write a program to replace all occurrences of a specific character with another character.
48. Write a program to remove all occurrences of a specified character from a string.
49. Write a program to print all characters located at even positions in a string.
50. Write a program to print all characters located at odd positions in a string.

Array Problems (51–100)

51. Write a program to find the largest element present in an integer array.
52. Write a program to find the smallest element present in an integer array.
53. Write a program to calculate the sum of all elements in an array.
54. Write a program to calculate the average value of all elements in an array.
55. Write a program to find the second largest element in an array.
56. Write a program to find the second smallest element in an array.
57. Write a program to reverse all elements of an array.
58. Write a program to display array elements in reverse order without modifying the array.
59. Write a program to sort an array in ascending order.
60. Write a program to sort an array in descending order.
61. Write a program to search for an element using linear search and display its index.
62. Write a program to search for an element using binary search.
63. Write a program to count the total number of even elements in an array.
64. Write a program to count the total number of odd elements in an array.
65. Write a program to determine how many times a given element appears in an array.
66. Write a program to find the frequency of every element in an array.
67. Write a program to identify and display duplicate elements in an array.
68. Write a program to remove duplicate values from an array.
69. Write a program to copy all elements from one array into another array.

70. Write a program to merge two arrays into a single array.
71. Write a program to find common elements present in two arrays.
72. Write a program to find elements that are unique to each of two arrays.
73. Write a program to find the intersection of two arrays.
74. Write a program to find the union of two arrays.
75. Write a program to rotate an array one position to the left.
76. Write a program to rotate an array one position to the right.
77. Write a program to move all zero elements to the end while maintaining the order of other elements.
78. Write a program to move all zero elements to the beginning of the array.
79. Write a program to count the number of positive and negative elements in an array.
80. Write a program to separate even and odd numbers into different groups.
81. Write a program to find the maximum value in an array.
82. Write a program to find the minimum value in an array.
83. Write a program to check whether an array is sorted in ascending order.
84. Write a program to check whether an array is sorted in descending order.
85. Write a program to find the index position of a given element in an array.
86. Write a program to insert an element at a specified position in an array.
87. Write a program to delete an element from a specified position in an array.
88. Write a program to replace all occurrences of a given value with another value.
89. Write a program to print elements located at even indices of an array.
90. Write a program to print elements located at odd indices of an array.
91. Write a program to find the first occurrence of a given element in an array.
92. Write a program to find the last occurrence of a given element in an array.
93. Write a program to count the occurrences of a particular element in an array.
94. Write a program to display all unique elements present in an array.
95. Write a program to traverse and print array elements using a for-each loop.
96. Write a program to create and display a two-dimensional array (matrix).
97. Write a program to calculate the sum of all elements in a two-dimensional array.

98. Write a program to find the largest element in a two-dimensional array.
99. Write a program to find the smallest element in a two-dimensional array.
100. Write a program to find the transpose of a matrix and display the resulting matrix.